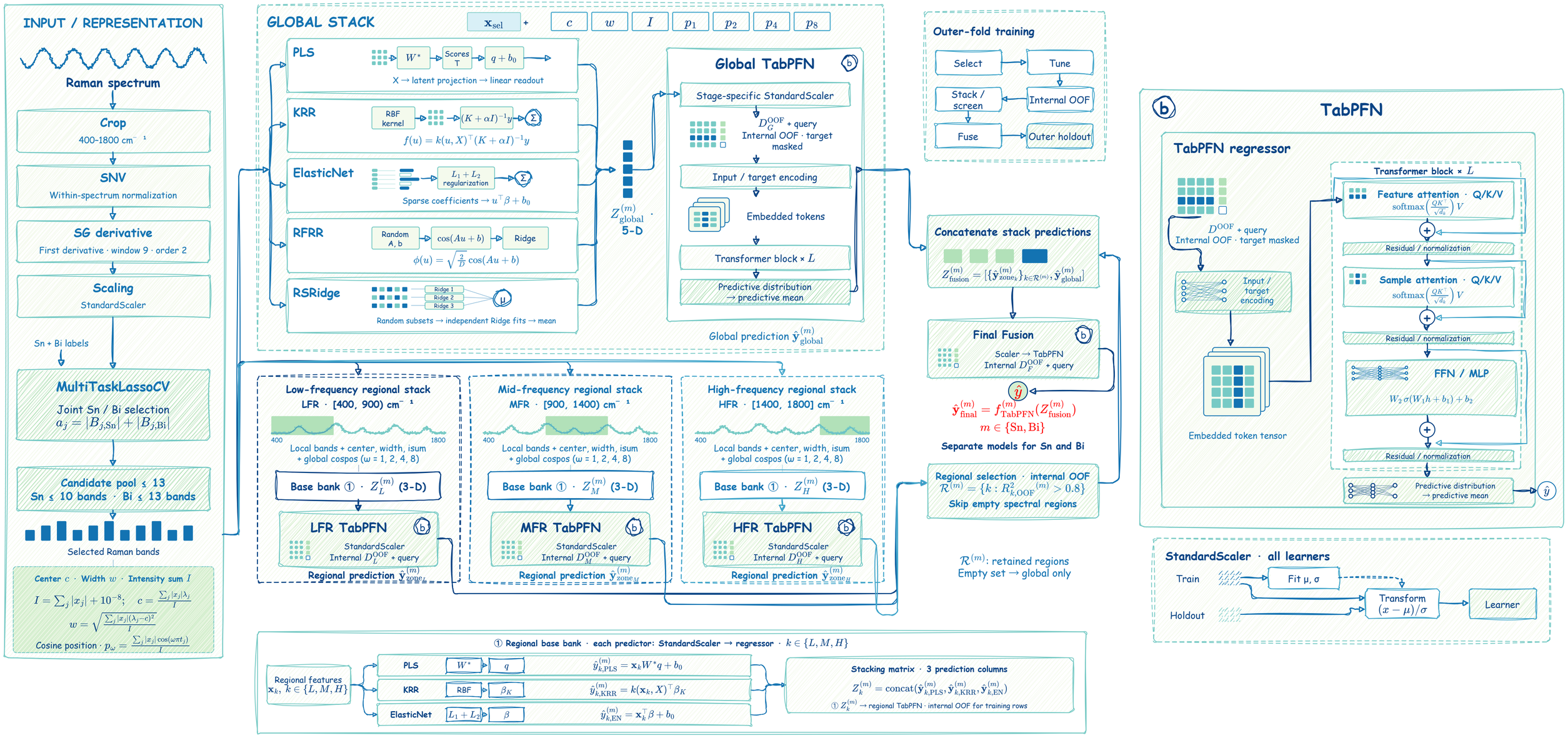

INPUT / REPRESENTATION
Raman spectrum
Crop
400–1800 cm⁻¹
SNV
Within-spectrum normalization
SG derivative
First derivative · window 9 · order 2
Scaling
StandardScaler
Sn + Bi labels
MultiTaskLassoCV
Joint Sn / Bi selection
$a_j = |B_{j,\mathrm{Sn}}| + |B_{j,\mathrm{Bi}}|$
Candidate pool ≤ 13
Sn ≤ 10 bands · Bi ≤ 13 bands
Selected Raman bands
Center c · Width w · Intensity sum I
$I = \sum_j |x_j| + 10^{-8}; \quad c = \frac{\sum_j |x_j| \lambda_j}{I}$
$w = \sqrt{\frac{\sum_j |x_j| (\lambda_j - c)^2}{I}}$
Cosine position · $p_\omega = \frac{\sum_j |x_j| \cos(\omega t_j)}{I}$
GLOBAL STACK
$\mathbf{x}_{\mathrm{sel}}$ + c w I p_1 p_2 p_4 p_8
PLS
W^* Scores T $q + b_0$
X → latent projection → linear readout
KRR
RBF kernel
$(K + \alpha I)^{-1} y$
$f(u) = k(u, X)^\top (K + \alpha I)^{-1} y$
ElasticNet
$L_1 + L_2$ regularization
Sparse coefficients → $u^\top \beta + b_0$
RFRR
Random A, b
$\cos(Au + b)$
Ridge
$\phi(u) = \sqrt{\frac{2}{D}} \cos(Au + b)$
RSRidge
Ridge 1
Ridge 2
Ridge 3
μ
Random subsets → independent Ridge fits → mean
$Z^{(m)}_{\mathrm{global}}$ 5-D
Global TabPFN
Stage-specific StandardScaler
D^{OOF}_G + query
Internal OOF · target masked
Input / target encoding
Embedded tokens
Transformer block × L
Predictive distribution → predictive mean
Global prediction $\hat{\mathbf{y}}^{(m)}_{\mathrm{global}}$
Low-frequency regional stack
LFR · [400, 900) cm⁻¹
Mid-frequency regional stack
MFR · [900, 1400) cm⁻¹
High-frequency regional stack
HFR · [1400, 1800] cm⁻¹
400
1800
Local bands + center, width, isum + global cospos (ω = 1, 2, 4, 8)
Base bank ① · $Z^{(m)}_L$ (3-D)
Base bank ① · $Z^{(m)}_M$ (3-D)
Base bank ① · $Z^{(m)}_H$ (3-D)
LFR TabPFN
MFR TabPFN
HFR TabPFN
StandardScaler
Internal D^{OOF}_L + query
Internal D^{OOF}_M + query
Internal D^{OOF}_H + query
Regional prediction $\hat{\mathbf{y}}^{(m)}_{\mathrm{zone}_L}$
Regional prediction $\hat{\mathbf{y}}^{(m)}_{\mathrm{zone}_M}$
Regional prediction $\hat{\mathbf{y}}^{(m)}_{\mathrm{zone}_H}$
① Regional base bank · each predictor: StandardScaler → regressor · $k \in \{L, M, H\}$
Regional features $\mathbf{x}_k$, $k \in \{L, M, H\}$
PLS W^* q $\hat{y}^{(m)}_{k,\mathrm{PLS}} = \mathbf{x}_k W^* q + b_0$
KRR RBF β_K $\hat{y}^{(m)}_{k,\mathrm{KRR}} = k(\mathbf{x}_k, X)^\top \beta_K$
ElasticNet $L_1 + L_2$ β $\hat{y}^{(m)}_{k,\mathrm{EN}} = \mathbf{x}_k^\top \beta + b_0$
Stacking matrix · 3 prediction columns
$Z^{(m)}_k = \mathrm{concat}(\hat{\mathbf{y}}^{(m)}_{k,\mathrm{PLS}}, \hat{\mathbf{y}}^{(m)}_{k,\mathrm{KRR}}, \hat{\mathbf{y}}^{(m)}_{k,\mathrm{EN}})$
① $Z^{(m)}_k$ → regional TabPFN · internal OOF for training rows
Outer-fold training
Select
Tune
Stack / screen
Internal OOF
Fuse
Outer holdout
Concatenate stack predictions
$Z^{(m)}_{\mathrm{fusion}} = [\{\hat{\mathbf{y}}^{(m)}_{\mathrm{zone}_k}\}_{k \in \mathcal{R}^{(m)}}, \hat{\mathbf{y}}^{(m)}_{\mathrm{global}}]$
Final Fusion
Scaler → TabPFN
Internal D^{OOF}_F + query
$\hat{y}$
$\hat{\mathbf{y}}^{(m)}_{\mathrm{final}} = f^{(m)}_{\mathrm{TabPFN}}(Z^{(m)}_{\mathrm{fusion}})$
$m \in \{\mathrm{Sn}, \mathrm{Bi}\}$
Separate models for Sn and Bi
Regional selection · internal OOF
$\mathcal{R}^{(m)} = \{k : R^{2\,(m)}_{k,\mathrm{OOF}} > 0.8\}$
Skip empty spectral regions
$\mathcal{R}^{(m)}$: retained regions
Empty set → global only
TabPFN
TabPFN regressor
D^{OOF} + query
Internal OOF · target masked
Input / target encoding
Embedded token tensor
Transformer block × L
Feature attention · Q/K/V
$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$
Residual / normalization
Sample attention · Q/K/V
FFN / MLP
$W_2\, \sigma(W_1 h + b_1) + b_2$
Predictive distribution → predictive mean
StandardScaler · all learners
Train
Holdout
Fit μ, σ
Transform $(x - \mu)/\sigma$
Learner